happened to perform the experiment on a sunny day in an open field, the surrounding buildings or trees, or the ground under our feet, would provide the “stationary” reference to determine whether the water was spinning. And if we happened to perform this experiment while floating in outer space, we would invoke the distant stars as our stationary reference.

This leads to the following question. Might Newton have kicked the bucket aside with such ease that he skipped too quickly over the relative motion we are apt to invoke in real life, such as between the water and the laboratory, or the water and the earth, or the water and the fixed stars in the sky? Might it be that such relative motion *can* account for the shape of the water’s surface, eliminating the need to introduce the concept of absolute space? That was the line of questioning raised by Mach in the 1870s.

To understand Mach’s point more fully, imagine you’re floating in outer space, feeling calm, motionless, and weightless. You look out and you can see the distant stars, and they too appear to be perfectly stationary. (It’s a real Zen moment.) Just then, someone floats by, grabs hold of you, and sets you spinning around. You will notice two things. First, your arms and legs will feel pulled from your body and if you let them go they will splay outward. Second, as you gaze out toward the stars, they will no longer appear stationary. Instead, they will seem to be spinning in great circular arcs across the distant heavens. Your experience thus reveals a close association between feeling a force on your body and witnessing motion with respect to the distant stars. Hold this in mind as we try the experiment again but in a different environment.

Imagine now that you are immersed in the blackness of *completely* empty space: no stars, no galaxies, no planets, no air, nothing but total blackness. (A real existential moment.) This time, if you start spinning, will you feel it? Will your arms and legs feel pulled outward? Our experiences in day-to-day life lead us to answer yes: any time we change from not spinning (a state in which we feel nothing) to spinning, we feel the difference as our appendages are pulled outward. But the current example is unlike anything any of us has ever experienced. In the universe as we know it, there are always other material objects, either